



CAV Risk Detection Integration Plan

Extending EV Route Optimizer with Computer Vision Capabilities

● Current State: EVRouteApp Architecture

BigQuery Connection EXISTING

- Priority: 1
- Google Cloud authentication
 - EV charging point data loading
 - Dataset: EVs_Infrastructure
 - Service account JSON key management

Road Network EXISTING

- Priority: 10
- OpenStreetMap integration
 - Road network download via osmdata
 - Spatial processing with sf package
 - Graph creation with dodgr

Route Optimizer EXISTING

- Priority: 20
- Shortest path calculation
 - Charging station integration
 - Origin/destination selection
 - Optimal route through charging points

Route Map EXISTING

- Priority: 30
- Leaflet-based visualization
 - Interactive route display
 - Charging point markers
 - HTML widgets integration

● Computer Vision Project Components

Python/Jupyter Workflow

- Google Maps API
- Polyline Decoding
- Haversine Distance
- Street View API
- YOLO (Ultralytics)
- Streamlit + Folium
- Pandas + GeoPandas

Detection Capabilities

- ✓ Roundabouts
- ✓ Tunnels
- ✓ Lane Merges
- ✓ Junctions
- ✓ Curves
- ✓ Pedestrian Crossings
- ✓ Construction Zones
- ✓ Obscured Signage

Processing Pipeline

1. Route extraction from Google Maps
2. Polyline decoding
3. 50m interval resampling
4. Street View image capture
5. YOLO object detection
6. Risk classification
7. Interactive map visualization